PULLBACK OF A K-FLAT COMPLEX WITH FLAT TERMS IS AGAIN K-FLAT WITH FLAT TERMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both assertions: flatness of the pulled back terms, the reduction to truncations, and the induction using the auxiliary flatness result are all inside the counted span. The Xy-pic diagram is delivered as written.

Editorial note. Every labelled reference occurring inside the counted spans points to a chapter of the Project that is not part of this card's source tree. Each is carried in assembly.external_references as a non-load-bearing navigation pointer whose source location is the invocation inside the counted span, so no load-bearing mathematics is replaced by a stub.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex of O-modules on a ringed topos. Same chapter, the definition of K-flatness through acyclicity of total tensor products with acyclic complexes. Supplies the property transferred by the pullback and the form in which the proof tests it.
- Flatness of the pullback of a flat module. Modules on Sites, the result that the pullback of a flat O-module is a flat O'-module. Supplies the flatness of the terms f^*K^n, which is the first assertion of the statement.
- Truncations of a complex and the auxiliary flatness results for them. Same chapter, the behaviour of K-flatness and flatness of terms under truncation and under the passage to a quotient, as recorded in the auxiliary lemmas of this section. Supplies the induction over the truncations with which the proof checks the acyclicity conditions defining K-flatness.

Source reference. Original lemma 05VD: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3518–3518. Navigation pointer invoked at proof line 3518; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 077J: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3173–3173. Navigation pointer invoked at proof line 3528; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03EV: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3532–3532. Navigation pointer invoked at proof line 3532; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YQ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3120–3120. Navigation pointer invoked at proof line 3539; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YR: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3152–3152. Navigation pointer invoked at proof line 3540; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03EU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3541–3541. Navigation pointer invoked at proof line 3541; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03EY: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3542–3542. Navigation pointer invoked at proof line 3542; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G7B: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3097–3097. Navigation pointer invoked at proof line 3544; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G6R: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3558–3558. Navigation pointer invoked at proof line 3558; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G7C: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3385–3385. Navigation pointer invoked at proof line 3565; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ be a morphism of ringed topoi. Let $\mathcal{K}^\bullet$ be a K-flat complex of $\mathcal{O}$-modules whose terms are flat $\mathcal{O}$-modules. Then $f^*\mathcal{K}^\bullet$ is a K-flat complex of $\mathcal{O}'$-modules whose terms are flat $\mathcal{O}'$-modules.*

Proof. The terms $f^*\mathcal{K}^n$ are flat $\mathcal{O}'$-modules by Modules on Sites, Lemma 05VD. Choose a diagram

$$\begin{array}{ccccc} \mathcal{K}_1^\bullet & \longrightarrow & \mathcal{K}_2^\bullet & \longrightarrow & \ldots \\ \downarrow & & \downarrow & & \\ \tau_{\leq 1}\mathcal{K}^\bullet & \longrightarrow & \tau_{\leq 2}\mathcal{K}^\bullet & \longrightarrow & \ldots \end{array}$$

as in Lemma 077J. We will use all of the properties stated in the lemma without further mention. Each $\mathcal{K}_n^\bullet$ is a bounded above complex of flat modules, see Modules on Sites, Lemma 03EV. Consider the short exact sequence of complexes

$$0 \to \mathcal{M}^\bullet \to \operatorname{colim} \mathcal{K}_n^\bullet \to \mathcal{K}^\bullet \to 0$$

defining $\mathcal{M}^\bullet$. By Lemmas 06YQ and 06YR the complex $\operatorname{colim} \mathcal{K}_n^\bullet$ is K-flat and by Modules on Sites, Lemma 03EU it has flat terms. By Modules on Sites, Lemma 03EY $\mathcal{M}^\bullet$ has flat terms, by Lemma 0G7B $\mathcal{M}^\bullet$ is K-flat, and by the long exact cohomology sequence $\mathcal{M}^\bullet$ is acyclic (because the second arrow is a quasi-isomorphism). The pullback $f^*(\operatorname{colim} \mathcal{K}_n^\bullet) = \operatorname{colim} f^*\mathcal{K}_n^\bullet$ is a colimit of bounded below complexes of flat $\mathcal{O}'$-modules and hence is K-flat (by the same lemmas as above). The pullback of our short exact sequence

$$0 \to f^*\mathcal{M}^\bullet \to f^*(\operatorname{colim} \mathcal{K}_n^\bullet) \to f^*\mathcal{K}^\bullet \to 0$$

is a short exact sequence of complexes by Modules on Sites, Lemma 0G6R. Hence by Lemma 0G7B it suffices to show that $f^*\mathcal{M}^\bullet$ is K-flat. This reduces us to the case discussed in the next paragraph.

Assume $\mathcal{K}^\bullet$ is acyclic as well as K-flat and with flat terms. Then Lemma 0G7C guarantees that all terms of $\tau_{\leq n}\mathcal{K}^\bullet$ are flat $\mathcal{O}$-modules. We choose a diagram as above and we will use all the properties proven above for this diagram. Denote $\mathcal{M}_n^\bullet$ the kernel of the map of complexes $\mathcal{K}_n^\bullet \to \tau_{\leq n}\mathcal{K}^\bullet$ so that we have short exact sequences of complexes

$$0 \to \mathcal{M}_n^\bullet \to \mathcal{K}_n^\bullet \to \tau_{\leq n}\mathcal{K}^\bullet \to 0$$

By Modules on Sites, Lemma 03EY we see that the terms of the complex $\mathcal{M}_n^\bullet$ are flat. Hence we see that $\mathcal{M} = \operatorname{colim} \mathcal{M}_n^\bullet$ is a filtered colimit of bounded below complexes of flat modules in this case. Thus $f^*\mathcal{M}^\bullet$ is K-flat (same argument as above) and we win. □